

Lab 8: Support Vector Machine (SVM)

Objective:

The purpose of this lab is to study the Support Vector Machine (SVM). Students will:

- Use two datasets to understand SVM classification.
- Apply preprocessing methods including standardization and PCA-based visualization.
- Utilize GridSearchCV() and Nested Cross Validation for tuning and evaluation.
- Display visualizations directly in the notebook for all comparisons, including PCA projections, support vectors, learning curves, confusion matrices, and decision boundaries.

Software Requirements:

- 1) Python
- 2) scikit-learn pandas
- 3) numpy
- 4) Matplotlib

Dataset:

sklearn.datasets: (1) The iris dataset; (2) The wine dataset.

Lab Instructions

1. Import necessary libraries.

2. Data Preparation, Visualization, Pre-processing, and Train-Test Split (10 pts)

A. Data Preparation (5 pts)

Load the two datasets and perform an initial data exploration. Specifically:

- Display the shape of each dataset.
- Print the first few rows to inspect sample entries.
- Examine the data structure and check for missing values.
- Summarize the statistical properties of numerical features (e.g., mean, standard deviation, min, max).

B. Train-Test Split (5 pts)

Split each dataset into training and testing subsets using a 70%-30% ratio.

Report the shapes of the resulting training and testing sets.

3. Scaling and PCA Visualization (15 pts)

After performing the train-test split: (10pt)

- Apply standardization using StandardScaler, fitting only on the training set and then transforming both training and testing sets.

- Use Principal Component Analysis (PCA) to visualize the data distribution before and after scaling.
- Plot the train set cumulative explained variance for PCA both before and after scaling.

Written Questions (5pt)

- Compare the PCA visualizations before and after scaling.
 - If there is little difference, explain why.
 - If there is a significant difference, explain the underlying reason.
- Compare the explained variance plots before and after scaling.
 - Discuss how scaling affects the variance distribution across principal components.

4. SVM Model Analysis, Optimization, and Evaluation (45pt)

Unless otherwise stated, all SVM models in this section should be trained using the standardized features.

A. Kernel comparison (10pt)

Compare the performance of different kernel functions in SVM.

- Train SVM models with the following kernels:
 - Linear
 - RBF
 - Polynomial
- Evaluate each model on the test set using accuracy.
- Visualize the comparison using a bar chart.

B. Hyperparameter tuning (10pt)

Optimize the SVM model using GridSearchCV.

- Define a parameter grid for:
 - C (regularization parameter)
 - gamma (kernel coefficient)
 - kernel
- Perform 5-fold cross-validation.
- Print out the best parameter combination.

C. Nested Cross Validation (10 pts)

Evaluate model performance using Nested Cross Validation. Use GridSearchCV as the inner loop and cross-validation as the outer loop. Compute the mean accuracy.

D. Training curve (5pt)

Compute training and validation scores for different training sizes. Plot the learning curve.

E. Evaluation (10pt)

Evaluate the optimized model on the test dataset. Predict labels on the test set. Compute accuracy and classification metrics. Visualize the confusion matrix.

5. Support vectors (30pt)

5.1 Decision boundary

- a. (10 pt) Apply Principal Component Analysis (PCA) to reduce high-dimensional data into a 2D feature space. Train Support Vector Machine (SVM) models with Linear, RBF, Polynomial, then visualize and compare their decision boundaries.
- b. (5 pt) Written Questions: Discuss how the choice of kernel affects model generalization and boundary smoothness.

5.2 Visualization

- a. (10pt) Plot the training data points in a 2D feature space. Highlight the support vectors using distinct markers. Display the total number of support vectors.
- b. (5pt) Written Questions : What are Support Vectors? (2) Interpretation of the Plot (3) What if the number of the support vectors is small, what if it is large?

Submission Requirement:

- 1) Complete task 1-5 in a jupyter notebook file, name your file as “name_student number.ipynb” and submit it through “Lab 8 Programming Assignment” on iSpace.
- 2) Please provide necessary outputs to support your assignment. (e.g. printing necessary information or visualizing through graphs and tables)

Deadline: 23:59, Apr.21, 2026, Plagiarism (cheating) = F